Food system model equations

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1 Soil equations

Soil health and quality index is modelled with the following differential equation:

$$\frac{dS}{dt} = (\alpha(S) - D(t))S(1 - S). \tag{1}$$

The quantity α represents the intrinsic recovery rate. When $\alpha > D(t)$, the soil is in recovery mode, and when $\alpha < D(t)$, indicates that the soil is subject to (net) degradation.

The intrinsic recovery rate can be modelled either as a constant or as

$$\alpha(S) = \frac{\alpha_{\max}}{1 + e^{-\rho(S - S_T)}}, \tag{2}$$

where $\alpha_{\max} > 0$ is the maximal recovery rate (for high S), and $\rho > 0$ controls the steepness of the nonlinear transition from small to large S, and S_T represents the inflection point where the transition (from slow to fast recovery rate) is steepest. As baseline values, we choose $\alpha_{\max} = 0.25$, $\rho = 50$, and $S_T = 0.2$.

2 Degradation term

The degradation term $D(t)$ depends on multiple factors. We assume that intensive farming is the primary driver of of soil degradation. In the absence of fertiliser use (and everything else being equal), we assume that farmland soil is degraded by an annual factor $\widehat{D}$. In this case, the overall degradation function is given by:

$$D(t) = \widehat{D}. \tag{3}$$

When there is no farming activity, $D(t)$ is set to be zero and the soil is left undisturbed to recover at the natural recovery rate of $\alpha(S)$.

We further assume that the use of fertilisers can either hamper the degradation rate or exacerbate it, depending on the type and quantity applied. Let $F(t) \geq 0$ denote the (annual) use of fertiliser, respectively, measured in $kg/(ha \cdot yr)$. Let the auxiliary function $D_f(F(t))$ denote the degradation rate of soil due to the application fertilisers, respectively. More formally, this function can be expressed as a linear combination of the degradation caused from fertiliser use:

$$D_f(F(t)) = (F(t) + 2)(F(t) - 1)^2, \tag{4}$$

Note: the critical point ($F = 1$) of synthetic dynamics is completely arbitrary and will need to be reviewed; the physically-irrelevant critical point ($F = -2$) merely controls the slope of proposed function. [ToDo: Identify the 'optimal' amount of fertiliser application $F = \widehat{F}$ and then choose the values of the two critical points such that the local minimum of $(F + 2)(F - 1)^2$ is $\widehat{F}$.]

The total degradation can then be expressed as follows:

$$D(t) = \widehat{D} + D_f(F(t)), \tag{5}$$

Note that a negative value of $D(t)$ corresponds to a positive recovery rate. Additionally, when $D(t)$ is positive but small enough, the net recovery rate, given by $\alpha(S) - D(t)$, remains positive.

3 Cover crops

Growing cover crops enhance the recovery rate by allowing the soil to increasing surface soil organic carbon and microbial biomass. To incorporate this impact, we introduce the binary input signal $\mathcal{B}(t) \in \{0, 1\}$, where $\mathcal{B}(t) = 1$ indicates that only cover crops are planted and $\mathcal{B}(t) = 0$ indicates that only cash crops (harvested crops) are planted. We $\mathcal{B}(t)$ model as a periodic function:

$$\mathcal{B}(t) = \begin{cases} 1, & \text{if } t \bmod \omega = \tau, \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

where $\tau, \omega \in \mathbb{Z}^{0,+}$, both measured in years, represent the duration of cover-crop phase and the length of intervention cycle, respectively. For example, when $\tau = 1$ and $\omega = 3$, cover crops are planted every three-year cycle for a period of one continuous year.

To ensure that degradation only occurs during cash-crop periods,, we modify the original degradation function (5) my multiplying the RHS by $(1 - \mathcal{B}(t))$:

$$D(t) = (1 - \mathcal{B}(t))(\widehat{D} + D_f(F(t))). \tag{7}$$

Thus, degradation is active only when $\mathcal{B}(t) = 0$, i.e., when cash crops are grown.

Fallow is farming practice of leaving farmland uncultivated for a period of time for a period of time to allow soil fertility to be restored. This practice can also be model using equation (6). Additionally, one may view cover crop planting as a subcategory of fallow.

4 Multiple lands

To model separate management practices, we partition the total land into N independent land areas. The sizes of the lands are denoted L_i, where $i \in 1, 2, ..., N$. The total land of the study area is given by $L = \sum_i^N L_i$. For simplicity, we assume that all land areas are equal and fixed in time.

We can also model the total farmed arable land using the following logistic equation:

$$\frac{dL}{dt} = \alpha L \left(1 - \frac{L}{K}\right), \tag{8}$$

where α is the (natural) growth rate and K represents the carrying capacity.

5 Yield

(Annual) yield rate $Y(t)$, measured in $tonnes/(ha \cdot yr)$, should be proportional to the amount of applied fertilisers and the condition of soil:

$$Y(t) = Y_f(F(t)) \cdot S(t)$$

The function $Y_f(F(t))$ should be a convex function with one global maximum.

This maximum should correspond to a higher fertiliser input compared to the fertilizer input that minimises soil degradation. Mathematically, this implies:

$$\arg\max_{F(t)\geq 0} Y_f(F(t)) > \arg\min_{F(t)\geq 0} D_f(F(t)) = \widehat{F}$$

6 Harvest

The (annual) total amount of crop harvest $H(t)$, measured in $tonnes/yr$, depends on the farming management and soil condition. More precisely, we model this variable as:

$$H(t) = Y(t)(1 - \mathcal{B}(t))L \tag{9}$$

The food production for a certain land i, measured in $tonnes$, is thus given by:

$$H_i(t) = Y_i(t)(1 - \mathcal{B}_i(t))L_i. \tag{10}$$

The (annual) grand total food production for all land is given by $H_T(t) = \sum_i^N H_i(t)$, measured in $tonnes/yr$.

7 Emissions and sequestration

Need to model GHG emissions from intensive farming and fertiliser use. Sequestration (negative emissions) can be computed as function of cover crops.

8 Food demand

To estimate the (annual) total food consumption (or demand), we use the following formula:

$$C(t) = \gamma P(t) \tag{11}$$

where the constant γ is the average personal energy demand (or food demand) measured in $kcal/capita \cdot yr$ or $kg/capita \cdot yr$, and $P(t)$ is the total population at time t.

9 Self-sufficiency ratio

Self-sufficiency ratio (SSR) is calculated as

$$\mathrm{SSR}(t) = \frac{100\ H_T(t)}{C(t)}. \tag{12}$$

The units of $H_T(t)$ and $C(t)$ need to be the same, either in $kcal/capita$ or $kg/capita$. SSR is a key indicator of food security.

10 Inflation, wages and income distribution

We define inflation index $I(t)$ as reference index to a certain year. For example, we could choose the year 2015 as the reference year and define $I(2025) = 1$. We model inflation index $I(t)$ using the following differential equation:

$$\frac{dI}{dt} = \delta_I I, \tag{13}$$

where the constant parameter δ_I is the year-on-year inflation rate.

Similarly, we define income index $W(t)$ as the (normalised) average income of the population. For example, we can set $W(2025) = 1$ to use the the (normalised) average income in 2025 as a reference. We model this index as follows:

$$\frac{dW}{dt} = \delta_W W, \tag{14}$$

where the constant parameter δ_W is the year-on-year increase (or decrease) in income.

Income encompasses many sources, such as wages, capital gains, social benefits, etc. While wages have been stagnating relative to inflation, capital gains and social benefits are subject to different dynamics. Furthermore, average income does not account for inequalities. Therefore, it is important to consider the entire distribution of income and how it evolves in time in comparison to inflation...

11 Inequality index

Inequality index can be calculated directly from income distribution. Alternatively, this index can be used to parameterise the income distribution.

12 Affordability index

Affordability index should account for income distribution, SSR, and subsidies.

13 Degradation term (Old)

The degradation term $D(t)$ depends on multiple factors. We assume that intensive farming is the primary driver of of soil degradation. In the absence of fertiliser use (and everything else being equal), we assume that farmland soil is degraded by an annual factor $\widehat{D}$. In this case, the overall degradation function is given by:

$$D(t) = \widehat{D}. \tag{15}$$

When there is no farming activity, $D(t)$ is set to be zero and the soil is left undisturbed to recover at the natural recovery rate of $\alpha(S)$.

We further assume that the use of fertilisers can either hamper the degradation rate or exacerbate it, depending on the type and quantity applied. Let $F_s(t)$ and $F_g(t)$ denote the (annual) use of synthetic and organic fertiliser, respectively, measured in $kg/(ha{\cdot}yr)$. Let the auxiliary function $D_f(F_s(t), F_g(t))$ denote the degradation rate of soil due to the application fertilisers, respectively. More formally, this function can be expressed as a linear combination of the degradation caused from synthetic and organic fertilizers:

$$D_f(F_s(t), F_g(t)) = (F_s + 2)(F_s - 1)^2 + e^{-rF_g}, \tag{16}$$

where r is the rate of exponential decay of the organic-induced soil degradation [I need to review this equation. Surely, this function cannot be monotonically decreasing.]. The implicit dependence on t on the RHS is dropped for notational clarity. Note: the critical point ($F_s = 1$) of synthetic dynamics is completely arbitrary and will need to be reviewed; the physically-irrelevant critical point ($F_s = -2$) merely controls the slope of proposed function. [ToDo: Identify the 'optimal' amount of fertiliser application $F_s = \widehat{F}_s$ and then choose the values of the two critical points such that the local minimum of $(F_s + 2)(F_s - 1)^2$ is $\widehat{F}_s$.]

The total degradation can then be expressed as follows:

$$D(t) = \widehat{D} + D_f(F_s(t), F_g(t)), \tag{17}$$

Note that a negative value of $D(t)$ corresponds to a positive recovery rate. Additionally, when $D(t)$ is positive but small enough, the net recovery rate, given by $\alpha(S) - D(t)$, remains positive.

References